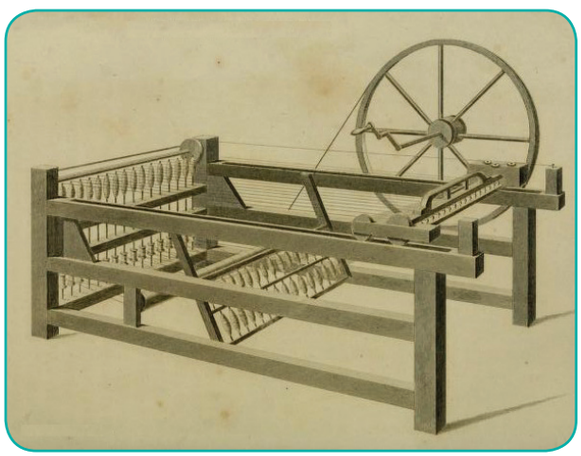
Cotton Industries: The first factories were established in the cotton industry. This became possible due to the invention of spinning jenny, flying shuttle, water frame and Crompton's Mule. Flying shuttle was invented by John Kay in 1733. Before this invention the thread in the shuttle in the weaver's hand had to be carried slowly across and through the other threads placed lengthwise, called the warp. The flying shuttle quickened this process and thus doubled the weaver's output. Spinning Jenny, invented in 1764 by James Hargreaves, could spin eight threads at the same time while in the traditional method only one thread could be spun. Water frame, developed in 1769 by Richard Arkwright, was able to spin 128 threads at a time. Crompton's Mule, a combination of Water Frame and Spinning Jenny, was invented by Samuel Crompton. It gave greater control over the weaving process and as a result, spinners could make many different types of yarn.



Spinning Jenny

In 1700, only 500 tons of cotton were imported by Britain. With innovations in spinning and weaving and the rise of factory production in textiles, the demand of raw cotton increased dramatically. By 1860, the country was importing 500,000 tons each year. By the early nineteenth century, Manchester, the centre of the British textile industry, had acquired the nickname "Cottonopolis".

Iron industries: Traditionally iron could be extracted from iron ore by heating (smelting) it. For this, a large quantum of charcoal was required which was obtained by burning firewood. Sources of coal had depleted by 1700 because of deforestation. Britain partially solved the problem, as about 1709, Abraham Darby a coal owner in Derbyshire, discovered that coke could be used for melting. The chief obstacle for the extraction of coal was the accumulation of water in the mines. What they found useful was a device, developed first by Thomas Newcomen in 1712, to pump the water from the coal mines. This was further improved by James Watt in 1769. He joined hands with an entrepreneur (Mathew Boulton) and together they produced more than 500 steam engines that were used to supply power to the new factories. The coming of power-driven machinery meant the rise of the factory system on a wide scale.



Newcomen Engine lifting coal from mines

Coke was smokeless and could produce more heat than charcoal. Due to this, iron industries were set up near coal mines. Due to the rapid production of iron many household objects such as spoons and pans were made of iron. Even the factories were built with strong iron girders.

Fascinated by the use of iron in the massive structures, the French in 1889 constructed the 324-metre-tall the Eiffel Tower in Paris.

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